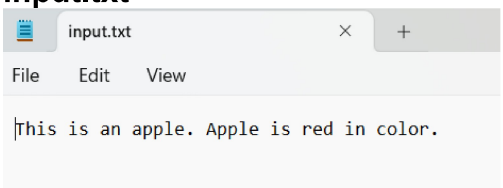
Name: Gaurang Vaghela
Rollno: BEAD-22561
CI Practical 3
Problem Statement: Design a distributed application using MapReduce under Hadoop for:

input.txt



a) Character counting in a given text file.

CharMapper.py

```
import sys
for line in sys.stdin:
    line = line.strip()
    for char in line:
        if char.strip() != "":
            print(f"{char}\t1")
```

CharReducer.py

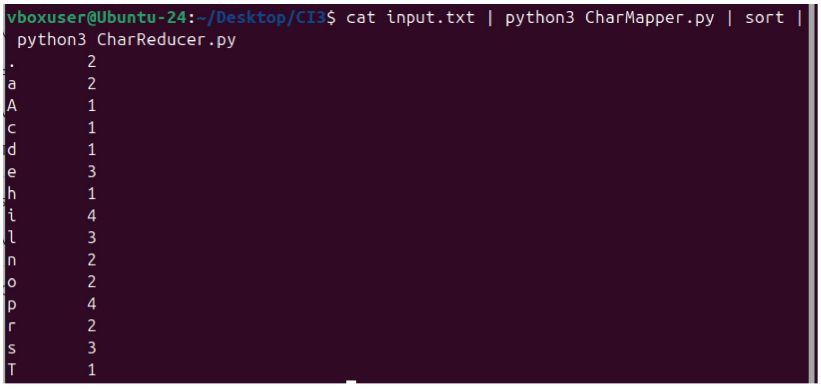
```
import sys

current_char = None
current_count = 0
for line in sys.stdin:
    char, count = line.strip().split('\t')
    count = int(count)

    if current_char == char:
        current_count += count
    else:
        if current_char:
            print(f"{current_char}\t{current_count}")
        current_char = char
        current_count = count

if current_char:
    print(f"{current_char}\t{current_count}")
```

OUTPUT:



```
vboxuser@Ubuntu-24:~/Desktop/CI3$ cat input.txt | python3 CharMapper.py | sort |
 python3 CharReducer.py
.       2
a       2
A       1
c       1
d       1
e       3
h       1
i       4
l       3
n       2
o       2
p       4
r       2
s       3
T       1
```

b) Counting no. of occurrences of every word in a given text file.

WordReducer.py

```
import sys

for line in sys.stdin:
    line = line.strip()
    words = line.split()

    for word in words:
        print(f"{word.lower()}\t1")
```

WordMapper.py

```
import sys

current_word = None
current_count = 0

for line in sys.stdin:
    word, count = line.strip().split('\t')
    count = int(count)

    if current_word == word:
        current_count += count
    else:
        if current_word:
            print(f"{current_word}\t{current_count}")
        current_word = word
        current_count = count

if current_word:
    print(f"{current_word}\t{current_count}")
```

OUTPUT:

```
vboxuser@Ubuntu-24:~/Desktop/CI3$ cat input.txt | python3 WordMapper.py | sort |
 python3 WordReducer.py
an      1
apple   1
apple.  1
color.  1
in      1
is      2
red     1
this    1
```